

Plan the story before you write it

One question • a figure storyboard • a topic-sentence outline

YOU LEAVE WITH

ONE committed question, a 3–5 figure storyboard with a claim under each figure, and your word budget.



Tutorial 4 of 8 • Semester 1

Today's plan

Goal: Leave with one question, a figure storyboard and a topic-sentence outline.

- 1 Methods: what r tells you; composite vs correlation vs slope
- 2 Methods: choosing (and testing) your region
- 3 Is your question good enough?
- 4 Storyboard, topic-sentence outline and word budget

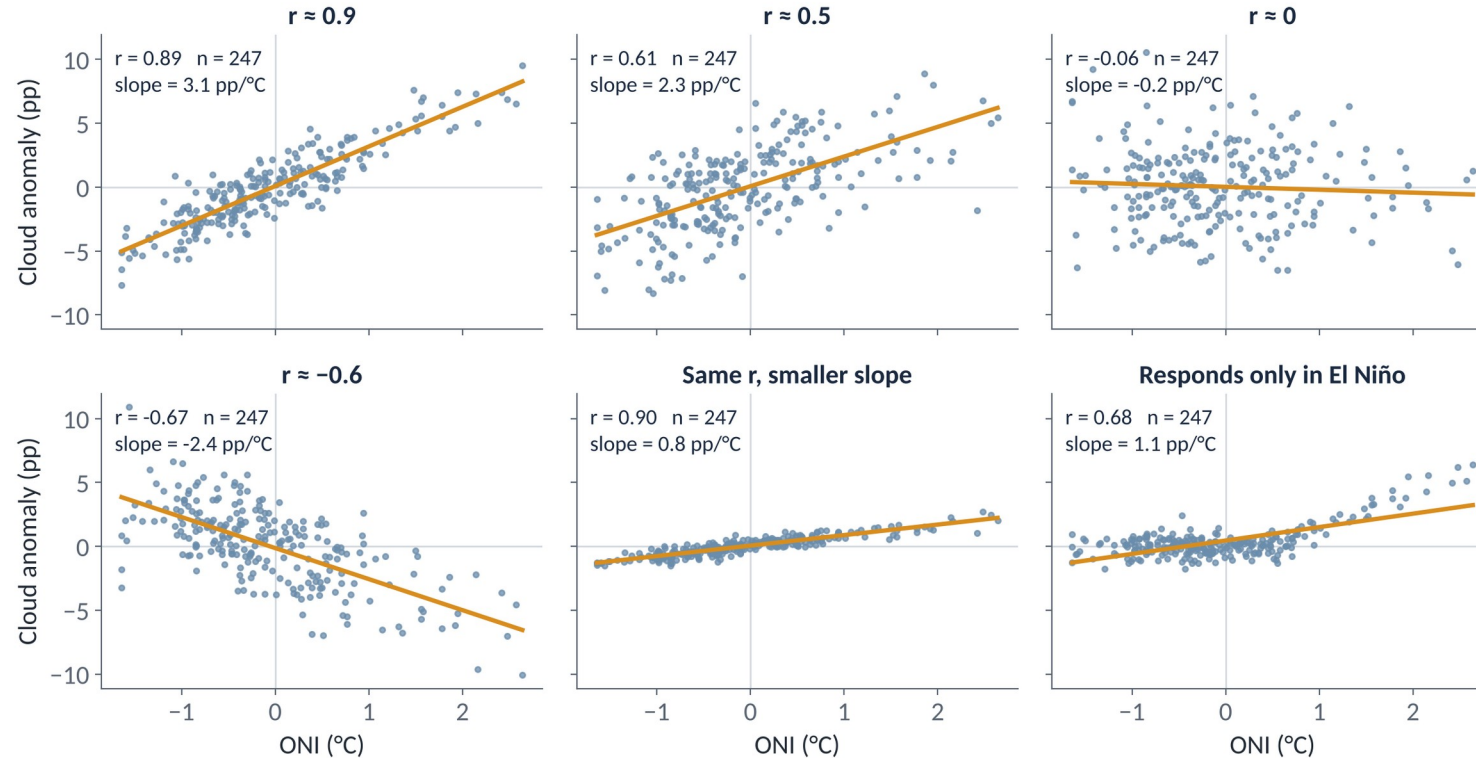
BRING / HAVE READY

- Task 3A opening paragraph
- Task 3B synthesis paragraph + references
- Task 3C: your **three candidate questions**
- Draft figures from notebook §4–5



METHODS

What r tells you, and what it doesn't



Real ONI (2005–2025) on the x-axis; synthetic cloud anomalies (not ERA5)

- **r**: how consistently two things vary together along a **straight line**, from -1 to $+1$. No units. $R^2 = r^2$: the fraction of variance a straight line explains ($r = 0.5 \rightarrow 25\%$).
- r is **not** the size of the effect; the **slope** is (pp per $^\circ\text{C}$ of ONI). Bottom middle: same r , about a quarter of the slope.
- r misses one-sided and curved responses. Bottom right only responds in El Niño, yet $r = 0.68$ looks respectable. And every r needs its n (Tutorial 5).

METHODS

Composite, correlation, slope: three questions

COMPOSITE

How different are El Niño and La Niña months, on average?

- Units: pp
- Uses only months beyond the threshold
- + simple; can show asymmetry (by phase)
- – few events; depends on the threshold

CORRELATION r

How consistently does cloud follow the ONI?

- No units, -1 to $+1$
- Uses every month
- + uses all the information
- – says nothing about size; straight lines only

REGRESSION SLOPE

How much does cloud change per $^{\circ}\text{C}$ of ONI?

- Units: pp per $^{\circ}\text{C}$
- Uses every month
- + gives the size of the effect
- – straight line; pulled by extremes (2015–16)

WHEN THEY DISAGREE, THAT'S INTERESTING

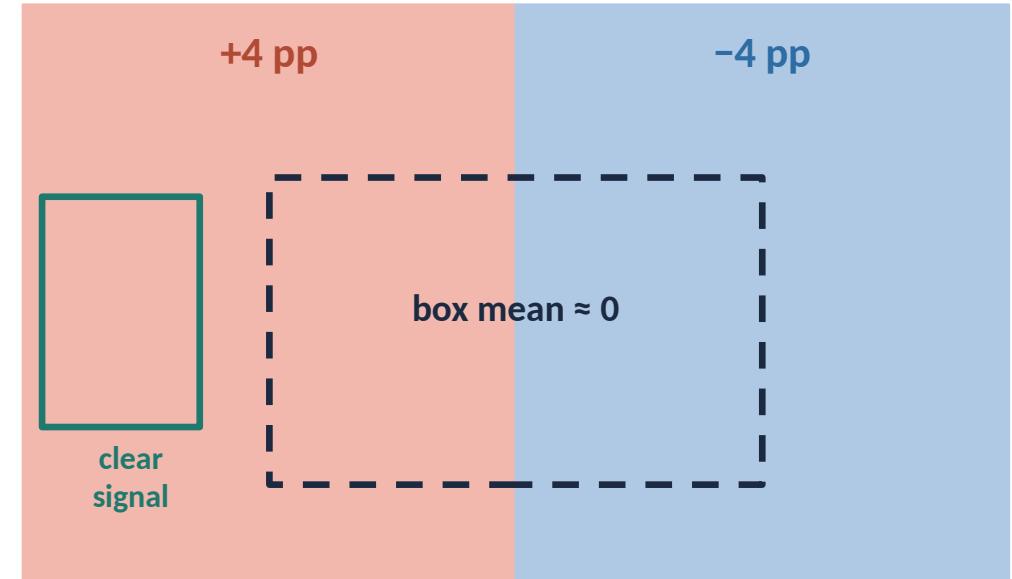
A big composite difference but a weak r usually means a few strong events or a one-sided response. Say so plainly. Schultz warns against claiming ‘remarkable similarity’ when, looked at closely, two results differ (§11.12).

METHODS

Choosing your region (and why it matters)

- A regional mean can **cancel out**: a box straddling positive and negative anomalies averages to almost nothing.
- Look at the maps first, then choose a box that sits on **one coherent** signal, for a **physical** reason.
- The notebook has 12 regions (e.g. 4 = Niño 4, 5°S–5°N, 160°E–150°W; 5 = Maritime Continent, 10°S–10°N, 100–150°E). You can add your own in REGIONS.
- **Sensitivity check**: shift the box by 5° and recompute. If the answer flips, say so.
- Write the **bounds** in your Methods.

WHY A BADLY PLACED BOX SEES NOTHING



Schematic

IN YOUR METHODS, THIS BECOMES

“The Maritime Continent region (MC; 10°S–10°N, 100–150°E) follows the notebook's predefined box.”

Is your question good enough?

- **Specific:** names a cloud type, region, season or timing.
- **Answerable:** with ERA5 + ONI and the notebook's tools.
- **Interesting:** it has a real 'however' (Tutorial 3).
- **Bounded:** 3–5 figures and about 650 words of Results.
- **Physical:** there is a mechanism you can discuss.
- Read your best candidate to a partner (20 s). They score it /5 against this list.

NARROWING A QUESTION, ONE RUNG AT A TIME

How does ENSO affect clouds?

too broad

How does ENSO affect high cloud?

+ cloud type

How does high cloud over the central Pacific respond to ENSO?

+ region

Does high cloud over the central Pacific respond to ENSO in the opposite sense to high cloud over the Maritime Continent, and which response is stronger?

+ a testable contrast

Storyboard: one claim under every figure

QUESTION: Does high cloud over the central Pacific respond to ENSO in the opposite sense to the Maritime Continent?

FIG. 1 • CONTEXT

El Niño – La Niña composite map of high cloud, both regions boxed

THIS FIGURE SHOWS THAT...

High-cloud anomalies have opposite signs in the two boxed regions.

FIG. 2 • TEST

Regional anomaly time series with ONI overlaid

THIS FIGURE SHOWS THAT...

Central Pacific high cloud rises in each El Niño; Maritime Continent high cloud falls.

FIG. 3 • QUANTIFY

Scatter of regional anomaly vs ONI, with r , n and slope

THIS FIGURE SHOWS THAT...

The Maritime Continent relationship is weaker ($r \approx []$) than the central Pacific one ($r \approx []$).

FIG. 4 • QUALIFY

Same, by season (or lag)

THIS FIGURE SHOWS THAT...

The Maritime Continent response is concentrated in [season].

Read **only the claims, left to right**. Do they answer the question in a sensible order? A claim you can't write means you don't yet know what that figure is for. A step with no figure means a gap. (Example claims are illustrative; yours come from your own analysis.)

Outline in topic sentences

- A paragraph has **one job**, announced by its **topic sentence** (Schultz ch. 8).
- Some authors outline by writing **every topic sentence first**. Read in order, they should tell the whole story.
- This is **assert-justify** from the proposal tutorials: state the point first, then the detail. Speed-readers get the message; careful readers read on.
- In Results, the topic sentence is usually the **finding**; the rest of the paragraph is the evidence.
- 2,000 words $\approx$ **15–18 paragraphs**. That's 15–18 topic sentences: about one side of A4.

EXAMPLE: RESULTS OUTLINE

- P1 High-cloud anomalies have opposite signs over the central Pacific and the Maritime Continent during El Niño (Fig. 1).
- P2 Central Pacific high cloud rises during El Niño and falls during La Niña (Fig. 2).
- P3 The Maritime Continent relationship is weaker than the central Pacific one (Fig. 3).
- P4 The Maritime Continent response is concentrated in [season] (Fig. 4).
- (illustrative; your sentences come from your figures)

Your 2,000-word budget

Introduction	Methods	Results	Discussion	Concl.	spare
350 ~3 ¶	300 2-3 ¶	650 4-6 ¶	450 3-4 ¶	150 1-2 ¶	100

HOW TO USE IT

- A guide, not a rule: but if Methods hits 600 words, something is wrong.
- Results gets the most words because it carries the evidence.
- Write your budget at the top of each section and check it when you finish.
- Over budget in a first draft is normal; you'll cut in January (Tutorial 8).

CHECK WITH YOUR TUTOR

- This budget assumes **figure captions, references and the title** are outside the 2,000 words. Your tutor will confirm what counts.
- Maximum is **2,000** for the final report. Being under is fine if the question is fully answered.
- Schultz: a paper 'should be as long as it needs to be, but no longer'.

Name things once: tag-phrases and lucid repetition

- From the proposal tutorials: give regions, methods and variables a short **tag-phrase**, define it once, then **never vary it**.
- Schultz calls deliberate repetition of key terms **lucid repetition**. It is how readers keep track.
- In English essays you were told to vary your words. In science, **varying a term implies a different thing**.
- Define abbreviations at first use: low-cloud cover (LCC), Oceanic Niño Index (ONI).

SPOT THE PROBLEM

“High cloud over the **central Pacific** rises during El Niño. In the **Niño 4 box**, the correlation is 0.8. Cloud **near the dateline** therefore...”

One region or three? A marker can't tell.

✓ Define “the central Pacific (CP; Niño 4 region, 5°S–5°N, 160°E–150°W)” once, then always write “CP”.

IN-CLASS ACTIVITY • 14 MIN

Commit, storyboard, outline

- 1 Choose **one** of your three candidate questions. Write it at the top of the storyboard page in your workbook.
- 2 Sketch **3–5 figure thumbnails** (rough is fine) and label each: context / test / quantify / qualify / compare.
- 3 Under each, write the **claim**: “This figure shows that...”. Use [] for numbers you don't have yet.
- 4 Turn the claims into **Results topic sentences** (P1, P2, ...).
- 5 Swap storyboards. Partner reads **only the claims** and answers the two questions on the right.

YOU LEAVE WITH

- One committed question
- A storyboard with a claim per figure
- Results topic sentences: the spine of your report

PAIR CHECK

- Do the claims, in order, **answer the question?**
- Which figure could be **removed** without weakening the answer? (If one can, remove it.)

Before Tutorial 5 • Thu 12 Nov

Two weeks (no tutorial on 5 Nov). About 2½ hours of writing; most of your time goes on the figures.

4A • notebook

Make the storyboard figures

- Adapt notebook analyses to **your** region / level / season
- Save **draft** versions of all 3–5 figures outside Binder
- Note the key numbers in your writing log

4B • 60 min

Methods draft, 250–300 words

- Use the Methods checklist in the workbook
- Data, period, region(s), ONI and thresholds, anomaly, weighting, statistics
- **Science, not code.** Bring it on 12 Nov for a peer check

4C • 40 min

Full topic-sentence outline

- Every paragraph of all five sections, one sentence each
- Mark the word budget beside each section
- Check: tags used consistently?

NOTEBOOK STRAND

- Analysis slides: **04 Choose your question** and **05 Build the evidence**.
- Optional §6 (autocorrelation / lag) only if your question is about timing.

READ • SCHULTZ (2009) ELOQUENT SCIENCE

- **Ch. 6** Brainstorm, outline, first draft
- **§4.8** Data and methods
- **§18.3** Picking thresholds